Lab 02 (CPCS 214)

Using simple instructions and branching in MIPS

1. Exercise

1. Write a program to compute the following equation `y = 2x+4z+5`. The output should be similar to the following

   ```
   Enter value of x: 5
   Enter value of z: 10
   y is 55
   ```

   Hint: Use the `sll` command for multiplication by 2 and 4. Use the command `li` to load a value directly to a register. For example, the command `li $t1, 5` loads the value 5 directly into the register $t1.

2. Write a program that prompts the user to enter a number, and displays "*This number is positive*" if the entered number is positive or should display "*This number is negative*" if it is negative. Here are two sample runs of the program:

   ```
   ## Sample run 1
   Enter an integer number: 5
   This number is positive
   ```

   ```
   ## Sample run 2
   Enter an integer number: -10
   This number is negative
   ```

   Hint : You may need to use a combination of some of the commands (`bne`, `beq`, `and`, `or`, `sll`, `j`, etc.) taught to you in class to complete this exercise.

3. Write a program that prompts the user to enter a number, and displays "*This number is even*" if the entered number is even otherwise displays "*This number is odd*" if the number entered is odd . Here is are two sample runs of the program

```
Enter an integer number: 15
This number is odd
```

```
Enter an integer number: 20
This number is even
```

Hint : You may need to use a combination of some of the commands (`bne` , `beq` , `and` , `or` , `sll` , `j` , etc.) taught to you in class to complete this exercise.

4. Write a program that reads two integer numbers and then prints the larger of the two numbers.

```
Enter two integer numbers: 5 10
10 is the larger number
```